A Lattice-Support Obstruction to Conjecture 00000000114

quantumtool

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Statement and interpretation

Let p_n be the n-th prime and let $g_n = p_{n+1} - p_n$. The conjecture says that, for each $k \geq 2$, the joint distribution of

$$(g_n, g_{n+1}, \ldots, g_{n+k-1})$$

converges to a vector of independent exponential random variables. The statement gives no normalization of the gaps. We use the usual empirical meaning of a limiting distribution for this deterministic sequence of vectors.

Contradiction

Fix $k = 2$. The empirical measures of the vectors are supported on $\mathbb{Z}^2_{>0}$, since every prime gap is a positive integer. Projecting onto the first coordinate is a continuous map, so weak convergence of the joint measures to a pair of independent exponential laws would imply weak convergence of the first-coordinate empirical measures to an exponential law.

Consider the interval $I = (1/4, 1/2)$. It contains no integer. Thus every empirical first-coordinate measure assigns I mass zero. In contrast, if X has an exponential distribution of any positive rate λ, then

$$\mathbb{P}(X \in I) = e^{-\lambda/4} - e^{-\lambda/2} > 0.$$

The endpoints of I have zero exponential mass, so I is a continuity set for the proposed limit. The portmanteau theorem would therefore force the empirical masses of I to converge to this strictly positive number, contradicting that they are identically zero.

Hence the conjecture as written is false: unnormalized integer gaps cannot have a continuous exponential limiting distribution. This argument requires no prime-gap estimates. A plausible repair would first divide each gap by a growing scale, commonly $\log p_n$; that is a different statement.

Lean formalization

The accompanying self-contained Lean project verifies the discrete arithmetic obstruction: after scaling by four, an integer in $(1/4, 1/2)$ would give a natural number n satisfying $1 < 4n < 2$, which is impossible. The weak-convergence, portmanteau, and continuous-mapping steps above are given in this paper and are not formalized in Lean.